GenAI: Test Planning

This module continues after **Requirement Analysis** in STLC and shows how **Generative AI supports Test Planning**.

Role of GenAI in Test Planning

Test Planning defines:

- Scope of testing
- Test strategy
- Resources
- Tools
- Schedule
- Risks

Generative AI helps by:

- Generating test plan templates
- Suggesting test strategies
- Identifying risks
- Estimating effort
- Recommending tools

How GenAI Supports Test Planning

Area	GenAI Contribution
Scope definition	Suggests modules to test
Strategy	Recommends manual/automation
Risk analysis	Identifies high-risk areas
Resource planning	Suggests roles
Schedule	Drafts timelines

Test Planning: Example

Project: Online Banking Application

Input Prompt

```
Create test plan for online banking app.
```

GenAI Output (Sample)

- Scope: Login, accounts, fund transfer
- Test Types: Functional, Security, Performance
- Tools: Selenium, JMeter
- Risks: Security breaches, downtime

Demo: Optimizing Test Planning for Healthcare Analytics Platform Using GitHub Copilot

Step 1: Create File

```
test_plan.md
```

Step 2: Type Comment

```
<!-- Create test plan for healthcare analytics platform -->
```

Step 3: Copilot Suggests

```
## Test Scope
- Data ingestion
- Dashboards
- Reports
- User management

## Test Types
- Functional
- Performance
- Security
- Compatibility

## Tools
- Selenium
- Postman
- JMeter
```

Step 4: Ask for Risks

```
<!-- Identify risks -->
```

Copilot suggests:

- Data accuracy issues
- HIPAA compliance risks
- Performance bottlenecks

Benefits of GenAI in Test Planning

- Faster plan creation
- Better coverage
- Reduced manual effort

Limitations

- Generic plans
- Needs customization

Quick Check

Q1. What is purpose of test planning?

Define testing approach

Q2. True/False

GenAI can suggest risks.
True

Q3. Best practice?

A) Use AI output as-is
B) Review and customize

B

Summary

GenAI accelerates test planning by:

- Drafting plans
- Identifying risks
- Suggesting strategies

But **human review is essential**.